Unit 4

Entrepreneurial Thinking

3 Slibraries · 15 framework sheets

Bird-in-Hand

SLIBRARY 10 · SHEET 1 / 5

Start with who you are, what you know, whom you know

PROFESSIONAL DEFINITION

Bird-in-Hand is Saraswathy Sarasvathy's first effectuation principle: entrepreneurs start with means at hand — their identity, knowledge and network — and let goals emerge from what those means make possible. The opposite of causal logic (which fixes goals and acquires means), means-first reasoning works under uncertainty by treating the entrepreneur's existing resources as the starting design constraint, not a limitation (Sarasvathy, 2001, 2008).

WHO DEVELOPED IT

Saras D. Sarasvathy, professor at the Darden School of the University of Virginia, in her 2001 *Academy of Management Review* paper drawing on think-aloud protocol studies of 27 expert entrepreneurs. The means triad — identity, knowledge, network — was canonised in *Effectuation* (2008).

WHY IT WAS DEVELOPED

Causal-rational textbook entrepreneurship taught founders to start with market analysis and goals — guidance that fails under genuine uncertainty when markets do not yet exist. Sarasvathy's research showed expert entrepreneurs systematically inverted this, starting with means; the principle codified that observed practice.

FIGURE 1 · THREE MEANS

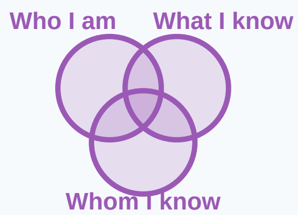

Figure 1. The three means at hand are the design constraints from which effectual ventures emerge.

HOW IT IS USED

Audit current means: who am I, what do I know, whom do I know? Generate venture options that those means make possible. Pick a direction not by market potential but by means-fit and stake-holder commitment.

WHEN IT IS USED

Use at venture inception under high uncertainty, in pivot decisions when market signals are weak, in second-act founder transitions, and when teaching entrepreneurship to first-time founders.

BY WHOM IT IS USED

First-time founders, accelerator programmes, MBA entrepreneurship courses, corporate intrapreneurs without external market mandate, and second-act career switchers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Drucker's (1985) entrepreneurship-by-opportunity-recognition assumed the opportunity existed and Kirzner's (1973) entrepreneurial-alertness theory was largely descriptive, Sarasvathy's contribution is the prescriptive means-first procedure. Building on Simon's (1959) bounded-rationality tradition, Bird-in-Hand makes effectual reasoning operational.

RELATED FRAMEWORKS IN THIS COURSE

Affordable Loss · Slibrary 10 | Lemonade · Slibrary 10 | Crazy Quilt · Slibrary 10 | Pilot-in-the-Plane · Slibrary 10

SOURCES (HARVARD REFERENCING STYLE)

Sarasvathy, S.D. (2001) 'Causation and Effectuation', *Academy of Management Review*, 26(2), pp. 243–263.
Sarasvathy, S.D. (2008) *Effectuation: Elements of Entrepreneurial Expertise*. Cheltenham: Edward Elgar.
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Kirzner, I.M. (1973) *Competition and Entrepreneurship*. Chicago: University of Chicago Press.

Affordable Loss

SLIBRARY 10 · SHEET 2 / 5

Commit only what you can afford to lose, not what you expect to gain

PROFESSIONAL DEFINITION

Affordable Loss is Sarasvathy's second effectuation principle: rather than committing resources based on expected returns (the causal-rational decision rule), entrepreneurs commit only what they can afford to lose. The principle reframes uncertainty: under conditions where expected return cannot be reliably calculated, downside management is the rational anchor. Affordable Loss bounds the failure cost of any single bet, making serial experimentation economically viable (Sarasvathy, 2001; Dew et al., 2009).

WHO DEVELOPED IT

Saras D. Sarasvathy in her 2001 *AMR* paper and 2008 book *Effectuation*; refined empirically with Dew, Read and Wiltbank in *Affordable Loss: Behavioural Economic Aspects* (2009). Conceptual roots in Knight's (1921) uncertainty/risk distinction.

WHY IT WAS DEVELOPED

Expected-utility decision theory required probability estimates that under genuine uncertainty did not exist. Founders following expected-return logic systematically over-committed and went bankrupt. Affordable Loss substituted a downside-bounded heuristic that worked precisely when expected-utility logic failed.

FIGURE 1 · EXPECTED RETURN VS AFFORDABLE LOSS

Causal	Effectual
Expected return	Affordable loss
Probabilities required	Downside cap
One large bet	Many small bets
Optimise	Survive & learn

Figure 1. Affordable Loss replaces probability-dependent calculations with downside-bounded experimentation.

HOW IT IS USED

Before any commitment, calculate what you can afford to lose — financially, reputationally, in time. Cap the bet at that level. Multiple small affordable bets across uncertain options dominate one large expected-return bet under genuine uncertainty.

WHEN IT IS USED

Use in early-venture investment decisions, in side-project commitments, in corporate-innovation portfolio sizing, and whenever expected-return calculation requires probabilities you cannot honestly estimate.

BY WHOM IT IS USED

Founders, angel investors, corporate-venture programme directors, and individual professionals making high-uncertainty career bets. Increasingly applied in personal finance and life-design literature.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Markowitz (1952) optimised expected return against variance and Knight (1921) distinguished risk from uncertainty, Affordable Loss's contribution is the operational decision rule for uncertainty cases — bounded downside replaces calculated expected value. Building on Simon's (1955) satisficing, it gives entrepreneurs a workable rule under irreducible unknowns.

RELATED FRAMEWORKS IN THIS COURSE

- Bird-in-Hand · Slibrary 10
- Knight's Risk vs Uncertainty · Slibrary 11
- Lemonade · Slibrary 10
- Pre-Mortem · Slibrary 11

SOURCES (HARVARD REFERENCING STYLE)

Sarasvathy, S.D. (2001) 'Causation and Effectuation', *Academy of Management Review*, 26(2), pp. 243–263.

Dew, N., Sarasathy, S., Read, S. and Wiltbank, R. (2009) 'Affordable Loss: Behavioural Economic Aspects of the Plunge Decision', *Strategic Entrepreneurship Journal*, 3(2), pp. 105–126.

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Sarasvathy, S.D. (2008) *Effectuation*. Cheltenham: Edward Elgar.

Lemonade

SLIBRARY 10 · SHEET 3 / 5

Treat surprises as resources — bad news as pivot opportunities

PROFESSIONAL DEFINITION

The Lemonade Principle is Sarasvathy's third effectuation principle: when life hands you lemons, make lemonade. Surprises — including unwelcome ones — are treated as data and as resources rather than as obstacles to plan-execution. The principle reframes the entrepreneur's relationship with the unexpected: causal logic minimises surprise (it perturbs the plan); effectual logic exploits surprise (it discloses what the plan could not have known) (Sarasvathy, 2001, 2008).

WHO DEVELOPED IT

Saras D. Sarasvathy in her 2001 *AMR* paper, derived from observed expert-entrepreneur protocol studies. Cultural roots in the proverbial American expression and conceptual links to Mintzberg's (1978) emergent-strategy tradition.

WHY IT WAS DEVELOPED

Plan-driven entrepreneurship treated surprises as failures requiring correction. Sarasvathy's research showed expert entrepreneurs systematically reframed surprises as information that the plan could not have anticipated — and that reframing was correlated with venture survival.

FIGURE 1 · SURPRISE → RESOURCE

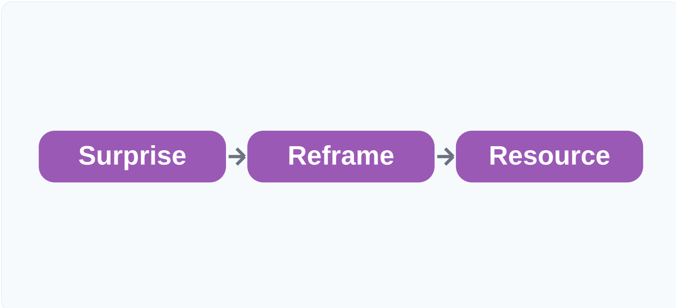

Figure 1. Effectual logic turns the unexpected into information; the same event is obstacle or resource depending on the frame.

HOW IT IS USED

When an unexpected event occurs, ask: what does this disclose that I could not have known before? What new resource or option does this create? Reframe the venture direction to incorporate the new information, rather than restoring the prior plan.

WHEN IT IS USED

Use whenever a venture encounters unexpected events — customer feedback, partner withdrawals, regulatory changes, technological shifts, market reactions — particularly during early-stage iteration.

BY WHOM IT IS USED

Founders navigating early-stage uncertainty, agile teams responding to user-test data, corporate-innovation labs reading market signals, and accelerator coaches helping teams interpret unexpected events.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Mintzberg (1978) identified emergent strategy as a phenomenon and Eisenhardt (1989) studied fast strategic decision-making, Lemonade's contribution is the prescriptive operating principle — surprise as resource. Building on Pasteur's (1854) prepared-mind aphorism, it operationalises serendipity as method.

RELATED FRAMEWORKS IN THIS COURSE

Bird-in-Hand · Slibrary 10 | Affordable Loss · Slibrary 10 | Crazy Quilt · Slibrary 10 | Pivot Types · Slibrary 16

SOURCES (HARVARD REFERENCING STYLE)

Sarasvathy, S.D. (2001) 'Causation and Effectuation', *Academy of Management Review*, 26(2), pp. 243–263.

Sarasvathy, S.D. (2008) *Effectuation*. Cheltenham: Edward Elgar.

Mintzberg, H. (1978) 'Patterns in Strategy Formation', *Management Science*, 24(9), pp. 934–948.

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Crazy Quilt

SLIBRARY 10 · SHEET 4 / 5

Build with self-selecting stakeholders — they bring new means

PROFESSIONAL DEFINITION

The Crazy Quilt Principle is Sarasvathy's fourth effectuation principle: entrepreneurs build the venture with whoever self-selects in — partners, customers, investors, advisers — rather than with a pre-specified team optimised against a fixed plan. Each new committed stakeholder brings their own means (knowledge, network, resources), expanding the venture's effectual basis. The metaphor of patchwork quilting captures the emergent, adaptive assembly process (Sarasvathy, 2001, 2008).

WHO DEVELOPED IT

Saras D. Sarasvathy in her 2001 *AMR* paper, derived from expert-entrepreneur protocol studies showing systematic preference for self-selecting stakeholder networks over optimised hire-against-plan teams. Conceptual roots in Granovetter's (1973) social-network theory.

WHY IT WAS DEVELOPED

Causal logic recruited team members against pre-specified job descriptions optimised for an assumed plan. But the plan was wrong (under uncertainty), and the team selection therefore was too. Sarasvathy's research showed expert entrepreneurs let the team emerge from who committed real stakes, exploiting the stakeholder's own evidence about whether to participate.

FIGURE 1 · SELF-SELECTING NETWORK

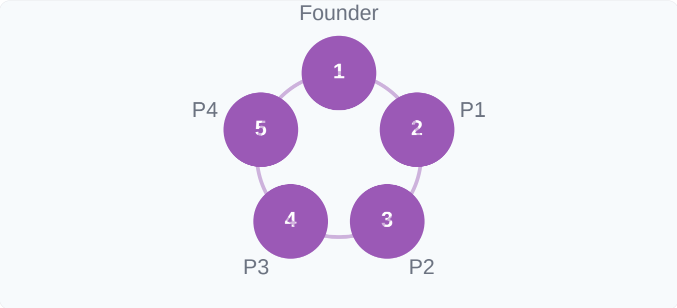

Figure 1. Each committed stakeholder brings their own means; the venture's effectual basis grows with the network.

HOW IT IS USED

When potential partners, customers or investors self-select toward the venture, ask: what means do they bring? How does their commitment expand what the venture can become? Co-design the next venture step with them rather than fitting them into an existing role.

WHEN IT IS USED

Use in early-stage venture team-building, in corporate intrapreneurship cohort formation, in social-venture coalition design, and whenever the team-formation question precedes the strategy-formation question.

BY WHOM IT IS USED

Founders, social entrepreneurs, accelerator programme designers, corporate-innovation cohort leads, and community-organising practitioners who build coalitions before campaigns.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Drucker's (1985) entrepreneurship treated team as plan-execution and traditional HR optimised hiring against role specifications, Crazy Quilt's contribution is the inversion — let the stakeholder set determine what the venture can become. Building on Granovetter's (1973) weak-tie research, it makes network-as-resource a strategic principle.

RELATED FRAMEWORKS IN THIS COURSE

- Bird-in-Hand · Slibrary 10
- Lemonade · Slibrary 10
- Pilot-in-the-Plane · Slibrary 10
- Founding Team Composition · Slibrary 16

SOURCES (HARVARD REFERENCING STYLE)

Sarasvathy, S.D. (2001) 'Causation and Effectuation', *Academy of Management Review*, 26(2), pp. 243–263.
Sarasvathy, S.D. (2008) *Effectuation*. Cheltenham: Edward Elgar.
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Pilot-in-the-Plane

SLIBRARY 10 · SHEET 5 / 5

The future is made, not predicted — focus on what you can control

PROFESSIONAL DEFINITION

The Pilot-in-the-Plane Principle is Sarasvathy's fifth and capstone effectuation principle: entrepreneurs treat the future as something *made* through human action rather than *predicted* through analysis. The metaphor positions the entrepreneur as actively flying the plane — adjusting controls in response to actual conditions — rather than as a passenger calculating arrival probabilities. This frames human agency, not market analysis, as the central source of venture outcomes (Sarasvathy, 2001, 2008).

WHO DEVELOPED IT

Saras D. Sarasvathy in her 2001 *AMR* paper and elaborated as the integrative principle in *Effectuation* (2008). Philosophical roots in pragmatist tradition (James, 1907; Dewey, 1922) and Frank Knight's (1921) entrepreneurship-as-judgment.

WHY IT WAS DEVELOPED

Predictive market analysis assumed the future was discoverable through better forecasting. Sarasvathy's research argued — drawing on both her empirical data and the philosophical pragmatist tradition — that the future is not pre-existing but constructed through entrepreneurial action, and that the methodological consequence is dramatic.

FIGURE 1 · MADE VS PREDICTED

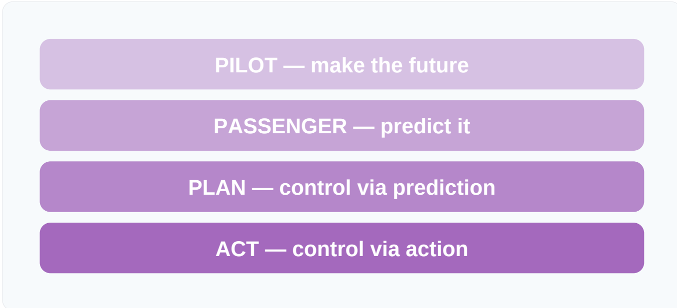

Figure 1. Entrepreneur as pilot adjusts to real conditions; passenger calculates arrival probabilities. The stance is the strategy.

HOW IT IS USED

Reject analyses that hinge on predicting market evolution. Instead ask: which futures could we and our partners actively bring about? Choose the one most aligned with means and commitments, then act to make it real rather than to forecast it.

WHEN IT IS USED

Use as the integrative posture under genuine uncertainty, in scenario-shaping workshops, in social-venture design, and whenever a team is paralysed by inability to predict markets accurately.

BY WHOM IT IS USED

Founders, social entrepreneurs, intrapreneurs in organisations facing existential strategic shifts, scenario-planning facilitators, and policy entrepreneurs designing futures rather than forecasting them.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Knight (1921) distinguished uncertainty from risk philosophically and Mintzberg (1978) described emergent strategy descriptively, Pilot-in-the-Plane's contribution is the actionable agency stance — entrepreneur as constructor. Building on the pragmatist tradition (James, 1907), it operationalises future-as-made into venture practice.

RELATED FRAMEWORKS IN THIS COURSE

- Bird-in-Hand · Slibrary 10
- Affordable Loss · Slibrary 10
- Knight's Risk vs Uncertainty · Slibrary 11
- Causal vs Effectual · Slibrary 12

SOURCES (HARVARD REFERENCING STYLE)

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Sarasvathy, S.D. (2008) *Effectuation*. Cheltenham: Edward Elgar.
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James, W. (1907) *Pragmatism: A New Name for Some Old Ways of Thinking*. New York: Longmans.

Knight's Risk vs Uncertainty

SLIBRARY 11 · SHEET 1 / 5

Risk: probabilities known. Uncertainty: unknowable. Different tools for each

PROFESSIONAL DEFINITION

Frank Knight's foundational 1921 distinction separates *risk* (situations where probabilities of outcomes are known or can be reliably estimated) from *uncertainty* (situations where probabilities themselves cannot be known). Risk is calculable and insurable; uncertainty is neither. Knight argued that genuine uncertainty — not risk — is the entrepreneur's domain, and that entrepreneurial profit is the reward for bearing uncertainty that no insurance market can price (Knight, 1921).

WHO DEVELOPED IT

Frank H. Knight, University of Chicago economist, in his doctoral dissertation published as *Risk, Uncertainty and Profit* (1921). The book founded the Chicago school of economics' approach to entrepreneurship and influenced both Austrian economics and Sarasvathy's effectuation theory.

WHY IT WAS DEVELOPED

Late-19th-century neoclassical economics treated all decision-making as expected-utility maximisation under known probabilities. Knight argued this collapsed into the theoretical equivalent of insurance — and that genuine entrepreneurship, by definition, dealt with situations no insurance could price.

FIGURE 1 · RISK VS UNCERTAINTY

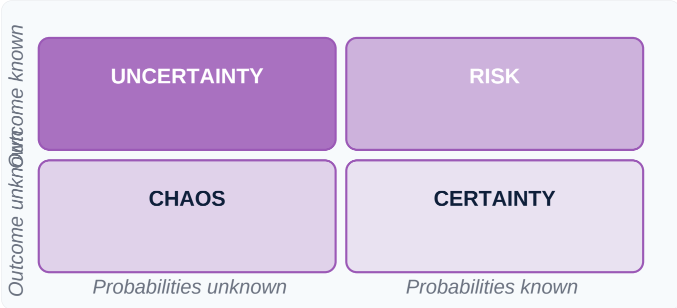

Figure 1. Risk admits expected-utility tools; uncertainty does not. Diagnosing which is which is the first strategic decision.

HOW IT IS USED

Diagnose each strategic situation: are the probabilities estimable from data, or are they fundamentally unknown? Risk situations admit expected-utility tools (Markowitz, decision trees, Monte Carlo). Uncertainty situations require effectual tools (Affordable Loss, Pilot-in-the-Plane, Pre-Mortem).

WHEN IT IS USED

Use at the start of any strategic-decision process — the diagnosis determines which toolkit applies. Use also in challenging quantitative analyses that disguise uncertainty as risk to give the appearance of rigour.

BY WHOM IT IS USED

Strategy practitioners, entrepreneurs, policy analysts, and finance professionals. Increasingly used in scenario planning and AI-policy contexts where both risk and uncertainty are present.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Bernoulli (1738) and Bayes (1763) developed probability theory for known-distribution cases, Knight's contribution is the categorical separation of probability-knowable from probability-unknowable cases. Building on the same year as Keynes's (1921) *A Treatise on Probability*, it gave entrepreneurship its theoretical home in the unknown-unknown.

RELATED FRAMEWORKS IN THIS COURSE

- Possibility Quadrant · Slibrary 11
- Real Options Lens · Slibrary 11
- Pre-Mortem · Slibrary 11
- Black Swan & Antifragility · Slibrary 11

SOURCES (HARVARD REFERENCING STYLE)

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Taleb, N.N. (2007) *The Black Swan: The Impact of the Highly Improbable*. New York: Random House.

Possibility Quadrant

SLIBRARY 11 · SHEET 2 / 5

Furr's 2×2 — map your uncertainty by certainty of cause and outcome

PROFESSIONAL DEFINITION

Nathan Furr's Possibility Quadrant is a 2×2 mapping of any situation against two axes: certainty of cause (do we know what works?) and certainty of outcome (do we know what we want?). The four quadrants prescribe different strategies: Causal (both known — risk management), Scenario Planning (cause uncertain), Real Options (outcome uncertain) and Effectual (both uncertain — act-and-learn). The frame integrates risk-management, scenario, options and effectual logics into one coherent decision tool (Furr and Furr, 2022).

WHO DEVELOPED IT

Nathan Furr, INSEAD professor, with Susannah Harmon Furr, in their 2022 book *The Upside of Uncertainty*. Synthesises insights from Sarasvathy (effectuation), Schoemaker (scenario planning), McGrath and MacMillan (real options), and Knight (risk).

WHY IT WAS DEVELOPED

Each existing uncertainty toolkit — risk-management, scenario planning, real options, effectuation — claimed universal applicability but worked best in specific conditions. The Possibility Quadrant integrates them as conditionally appropriate, prescribing which tool fits which uncertainty type.

FIGURE 1 · FOUR UNCERTAINTY QUADRANTS

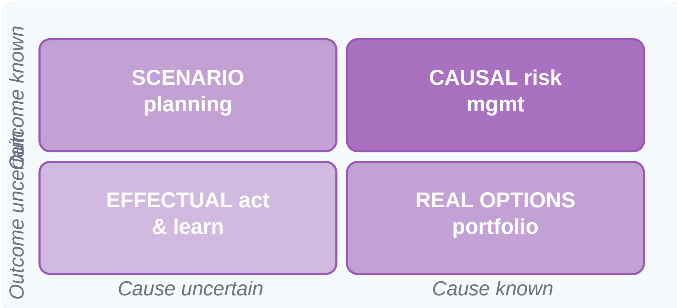

Figure 1. Each quadrant prescribes a different toolkit. The diagnosis precedes the technique. Adapted from Furr & Furr (2022).

HOW IT IS USED

Diagnose the situation against the two axes. Apply the toolkit appropriate to that quadrant. As certainty changes (typically increasing as a venture progresses), the situation may move quadrants — and the toolkit must shift with it.

WHEN IT IS USED

Use at the start of strategic decisions to determine which toolkit applies, in scenario-planning workshops, in venture-pivot reviews, and as a teaching frame for uncertainty management.

BY WHOM IT IS USED

Strategy and innovation leaders, venture-capital partners, policy entrepreneurs, and MBA students learning multiple uncertainty toolkits.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where each prior toolkit (Knight, 1921; Sarasvathy, 2001; Schoemaker, 1995; McGrath & MacMillan, 1995) operated independently, Furr's contribution is the integrative 2×2 that locates each toolkit in its appropriate condition. Building on Stacey's (1996) complexity matrix, it makes uncertainty-toolkit selection a deliberate strategic choice.

RELATED FRAMEWORKS IN THIS COURSE

- Knight's Risk vs Uncertainty · Slibrary 11
- Real Options Lens · Slibrary 11
- Pre-Mortem · Slibrary 11
- Causal vs Effectual · Slibrary 12

SOURCES (HARVARD REFERENCING STYLE)

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McGrath, R.G. and MacMillan, I.C. (1995) 'Discovery-Driven Planning', *Harvard Business Review*, 73(4), pp. 44–54.

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Real Options Lens

SLIBRARY 11 · SHEET 3 / 5

Uncertainty has value — stage investments to preserve options

PROFESSIONAL DEFINITION

The Real Options Lens applies financial-options thinking to strategic investments: each commitment is structured to preserve the right (but not the obligation) to expand, defer, contract or abandon based on future information. Under uncertainty, optionality has measurable value — staged commitments that resolve uncertainty before doubling down beat single large commitments that lock in early. The lens reframes uncertainty from threat to value source (McGrath and MacMillan, 1995; Trigeorgis, 1996).

WHO DEVELOPED IT

Stewart Myers (1977) coined "real options". Lenos Trigeorgis (1996) gave it a comprehensive academic treatment; Rita McGrath and Ian MacMillan (1995, 2009) translated it into management practice through *Discovery-Driven Planning* and *Discovery-Driven Growth*.

WHY IT WAS DEVELOPED

Discounted-cash-flow analysis under uncertainty systematically undervalued projects with optionality (e.g., R&D;, new-market entry) because it ignored the value of being able to expand or abandon. Real-options thinking made that value explicit, justifying staged investments that DCF would have rejected.

FIGURE 1 · STAGED OPTIONALITY

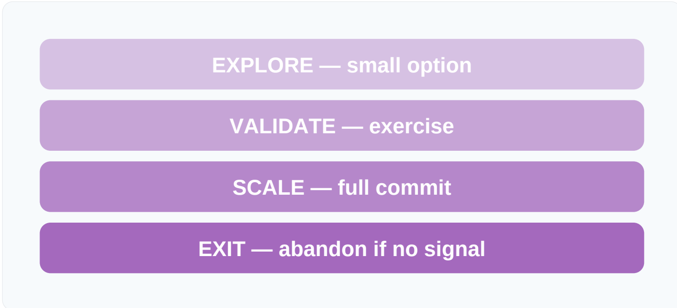

Figure 1. Each stage preserves the right but not the obligation to continue; uncertainty resolves before commitment escalates.

HOW IT IS USED

Decompose major commitments into stages. At each stage, structure the next commitment as a real option — exercisable if uncertainty resolves favourably. Value the option explicitly when comparing alternatives.

WHEN IT IS USED

Use in strategic-investment decisions under high uncertainty, in R&D; portfolio sizing, in venture-capital staged-funding design, and when DCF analysis appears to reject projects with high optionality value.

BY WHOM IT IS USED

Corporate-finance teams, strategy consultants, venture capitalists structuring staged rounds, and innovation portfolio managers balancing committed-vs-optional bets.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Black and Scholes (1973) priced financial options and Markowitz (1952) optimised portfolios under known distributions, Real Options' contribution is applying optionality logic to strategic — not financial — decisions. Building on Myers's (1977) initial insight, McGrath and MacMillan (1995) made it management-practical.

RELATED FRAMEWORKS IN THIS COURSE

- Knight's Risk vs Uncertainty · Slibrary 11
- Possibility Quadrant · Slibrary 11
- Pre-Mortem · Slibrary 11
- Discovery Interviews · Slibrary 16

SOURCES (HARVARD REFERENCING STYLE)

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Pre-Mortem

SLIBRARY 11 · SHEET 4 / 5

Imagine the project has failed — work backwards to causes now

PROFESSIONAL DEFINITION

The Pre-Mortem is a prospective version of the post-mortem developed by Gary Klein: before commitment, the team imagines that the project has already failed catastrophically and works backwards to write the autopsy explaining why. The frame inverts standard project planning — instead of forecasting success, the team is required to construct plausible failure narratives, surfacing risks that optimism bias and groupthink would have suppressed (Klein, 2007).

WHO DEVELOPED IT

Gary Klein, naturalistic-decision-making researcher, in his 2007 *Harvard Business Review* article and elaborated in *Streetlights and Shadows* (2009). Conceptual roots in Mitchell, Russo and Pennington's (1989) prospective hindsight research and Kahneman's (1973) availability-heuristic studies.

WHY IT WAS DEVELOPED

Project teams systematically under-predicted failure because the social environment of project planning rewarded optimism and punished raised concerns. Klein's pre-mortem reverses the social incentive — failure scenarios become the explicit assignment, making risk-raising team-positive rather than team-disruptive.

FIGURE 1 · PRE-MORTEM SEQUENCE

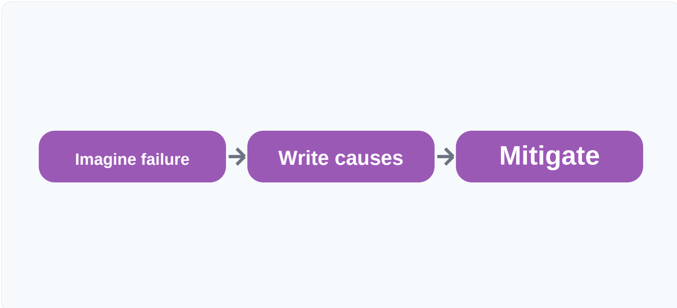

Figure 1. Prospective hindsight surfaces 30% more failure causes than forward-looking risk analysis (Mitchell et al., 1989).

HOW IT IS USED

Before committing, gather the team and announce: imagine 12 months from now this project has failed catastrophically. Each member silently writes the reasons. Discuss and consolidate. The findings inform mitigations, kill criteria or the decision to proceed.

WHEN IT IS USED

Use before committing to any major bet, in stage-gate reviews, in M&A; go/no-go decisions, in venture-capital investment-committee meetings, and as a counterweight to teams in optimistic momentum.

BY WHOM IT IS USED

Project sponsors, corporate-strategy teams, venture-capital investment committees, accelerator programme directors, and any decision-maker resisting the optimism bias of execution-focused teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where post-mortems documented failure after the fact and risk registers listed risks abstractly, Klein's contribution is the prospective hindsight technique — making failure narratives concrete before commitment. Building on Mitchell et al.'s (1989) research showing prospective hindsight produces 30% more failure causes, it operationalises that finding as a workshop ritual.

RELATED FRAMEWORKS IN THIS COURSE

Possibility Quadrant · Slibrary 11 | Real Options Lens · Slibrary 11 | 5 Whys · Slibrary 3 | Sprint Retrospective · Slibrary 6

SOURCES (HARVARD REFERENCING STYLE)

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Black Swan & Antifragility

SLIBRARY 11 · SHEET 5 / 5

Build systems that gain from disorder, not merely survive it

PROFESSIONAL DEFINITION

Nassim Nicholas Taleb's twin concepts frame extreme uncertainty: *Black Swans* are rare, high-impact, retrospectively rationalised events that statistical models systematically under-weight; *Antifragility* is the property of systems that gain — not merely survive — from volatility, stress and disorder. Antifragile systems benefit from small failures because they absorb information from each one and convex-positively to upside surprises (Taleb, 2007, 2012).

WHO DEVELOPED IT

Nassim Nicholas Taleb, scholar and former options trader, in *The Black Swan* (2007) and *Antifragile* (2012). Built on his earlier work *Fooled by Randomness* (2001) and a tradition of fat-tail statistics tracing to Mandelbrot (1963).

WHY IT WAS DEVELOPED

Mainstream risk management (VaR, normal-distribution Monte Carlo) systematically under-priced extreme events. Taleb argued — backed by both theoretical and empirical work — that the modal failure mode of complex systems was not the bell-curve middle but the fat-tail extremes, and that strategy must therefore engineer for antifragility, not robustness.

FIGURE 1 · THREE STRESS RESPONSES

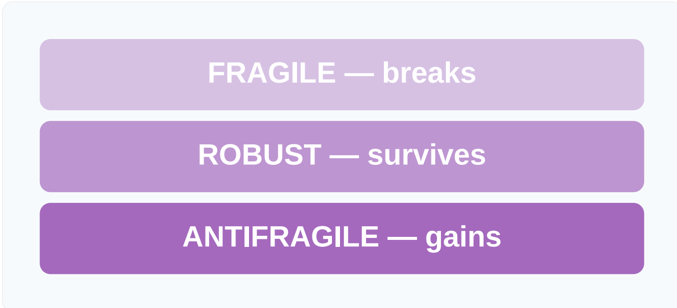

Figure 1. Antifragility goes beyond robustness; some systems benefit from the disorder others merely survive.

HOW IT IS USED

Audit each system for fragility (does it lose more from extreme stress than it gains from extreme calm?) versus antifragility (the opposite). Optionality, redundancy, small-bet portfolios and via-negativa interventions tilt systems toward antifragile.

WHEN IT IS USED

Use in risk-management redesign, in venture-portfolio construction, in critical-infrastructure planning, in personal financial design, and in strategic-resilience assessments after major shocks.

BY WHOM IT IS USED

Risk managers, portfolio investors, public-sector resilience leaders, founders designing for survival in uncertain markets, and policy analysts studying complex-system robustness.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Mandelbrot (1963) showed financial returns had fat tails and Knight (1921) distinguished risk from uncertainty, Taleb's contribution is the actionable design principle of antifragility — engineering for benefit-from-disorder, not just survival-of-disorder. Building on Hayek's (1945) decentralisation tradition, it gives strategists prescriptive guidance for irreducible uncertainty.

RELATED FRAMEWORKS IN THIS COURSE

- Knight's Risk vs Uncertainty · Slibrary 11
- Real Options Lens · Slibrary 11
- Pre-Mortem · Slibrary 11
- Affordable Loss · Slibrary 10

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Causal vs Effectual

SLIBRARY 12 · SHEET 1 / 5

Goal-first vs means-first — experts switch deliberately

PROFESSIONAL DEFINITION

Sarasvathy's foundational distinction separates two logics of decision-making: *causal* reasoning starts with a goal and acquires the means to achieve it (the textbook MBA approach); *effectual* reasoning starts with available means and lets goals emerge. Empirical studies of expert entrepreneurs show that they do not abandon causal logic — they switch between the two deliberately, applying causal thinking to known-domain problems and effectual thinking to genuinely uncertain ones (Sarasvathy, 2001; Read et al., 2009).

WHO DEVELOPED IT

Saras D. Sarasvathy in her foundational 2001 *Academy of Management Review* paper, drawing on protocol analyses of 27 expert entrepreneurs. The deliberate-switching finding was empirically confirmed in subsequent work by Read, Dew, Wiltbank and Sarasvathy.

WHY IT WAS DEVELOPED

Existing entrepreneurship education taught only causal logic — start with the goal, write a business plan, raise capital, execute. Sarasvathy's empirical research showed expert entrepreneurs systematically used a different (effectual) logic in early-stage uncertainty, and that the difference was correlated with venture survival.

FIGURE 1 · TWO LOGICS

Causal	Effectual
Goal first	Means first
Predict future	Make future
Expected return	Affordable loss
Plan	Co-create

Figure 1. Experts switch deliberately by domain; the toolkit follows the diagnosis.

HOW IT IS USED

Diagnose each decision: is the domain knowable enough that goals and probabilities can be reliably specified? If yes, apply causal tools; if no, switch to effectual tools (Bird-in-Hand, Affordable Loss, Lemonade).

WHEN IT IS USED

Use as the meta-question at every strategic decision point. Particularly important in early-stage ventures, pivot decisions, and corporate-innovation projects entering unknown markets.

BY WHOM IT IS USED

Founders, venture capitalists evaluating early-stage decision-making, MBA students learning to alternate between toolkits, and corporate-innovation teams shifting between exploration and exploitation modes.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where causal-rational decision theory (Simon, 1957) and bounded-rationality research (Cyert and March, 1963) treated all decisions as one species, Sarasvathy's contribution is the empirical demonstration that experts deliberately switch logics by domain. Building on Knight's (1921) risk-uncertainty distinction, it gives the switch a theoretical home.

RELATED FRAMEWORKS IN THIS COURSE

- Bird-in-Hand · Slibrary 10
- Knight's Risk vs Uncertainty · Slibrary 11
- Pilot-in-the-Plane · Slibrary 10
- Possibility Quadrant · Slibrary 11

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Opportunity Recognition

SLIBRARY 12 · SHEET 2 / 5

Discovery · Creation · Enactment — three theories of opportunity

PROFESSIONAL DEFINITION

The Opportunity Recognition literature distinguishes three theories of where entrepreneurial opportunities come from: the *Discovery* view (opportunities exist objectively, awaiting alert entrepreneurs — Kirzner, 1973), the *Creation* view (opportunities are constructed through entrepreneurial action — Sarasvathy, 2008), and the *Enactment* view (opportunities and entrepreneurs co-emerge — Alvarez and Barney, 2007). The three theories prescribe different strategies; choosing the right frame matters more than picking sides (Alvarez and Barney, 2007).

WHO DEVELOPED IT

Israel Kirzner (1973) articulated discovery; Saras Sarasvathy (2001, 2008) championed creation; Sharon Alvarez and Jay Barney (2007) named the enactment frame and synthesised the three traditions in their *Strategic Entrepreneurship Journal* paper.

WHY IT WAS DEVELOPED

Entrepreneurship research had two conflicting traditions — Austrian discovery and effectual creation — that often spoke past each other. Alvarez and Barney's typology resolved the dispute by showing that each theory was correct for different opportunity types, not universally.

FIGURE 1 · THREE THEORIES

Figure 1. Three theories of opportunity origin; each prescribes a different strategy in a different uncertainty regime.

HOW IT IS USED

Diagnose each opportunity: does it pre-exist (discovery, in known markets), or is it constructed through founder action (creation, in uncertain markets), or does it emerge through interaction (enactment, in network-driven contexts)? Apply the matching strategy.

WHEN IT IS USED

Use when teams debate where opportunities come from, in early-venture-design workshops, in academic entrepreneurship courses, and when reconciling different schools of entrepreneurship thinking.

BY WHOM IT IS USED

Entrepreneurship researchers, MBA professors, founders developing venture theses, and policy analysts designing entrepreneurial-ecosystem interventions.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Kirzner (1973) and Sarasvathy (2001) each argued for one theory, Alvarez and Barney's contribution is the typology that locates each in its appropriate condition. Building on Eckhardt and Shane's (2003) opportunity literature review, it makes opportunity-theory selection a deliberate strategic choice.

RELATED FRAMEWORKS IN THIS COURSE

- Causal vs Effectual · Slibrary 12
- Bird-in-Hand · Slibrary 10
- Possibility Quadrant · Slibrary 11
- JTBD Canvas · Slibrary 13

SOURCES (HARVARD REFERENCING STYLE)

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Bias for Action

SLIBRARY 12 · SHEET 3 / 5

Default to shipping — small imperfect action beats large perfect plan

PROFESSIONAL DEFINITION

Bias for Action is a venture-mindset principle: when in doubt, default to shipping over planning. Small imperfect actions generate evidence; large perfect plans generate confidence without evidence. Under uncertainty, the marginal value of one more day of action exceeds one more day of analysis — a finding empirically associated with venture survival across multiple studies. The bias is engineered, not natural; teams must explicitly counteract analysis-paralysis (Peters and Waterman, 1982; Eisenhardt, 1989).

WHO DEVELOPED IT

Tom Peters and Robert Waterman codified "a bias for action" as the first of eight attributes of excellent companies in *In Search of Excellence* (1982); Eisenhardt's (1989) fast-strategic-decision research operationalised it empirically. Cultural roots in Toyota's *genchi genbutsu* (go-and-see) discipline (Liker, 2004).

WHY IT WAS DEVELOPED

MBA-trained managers over-invested in analysis under uncertainty — analysis itself was the comfortable activity. Peters and Waterman argued that excellent firms inverted the default: ship first, refine later. Eisenhardt's research showed faster decision-making correlated with firm performance even when controlling for decision quality.

FIGURE 1 · ACTION VS ANALYSIS

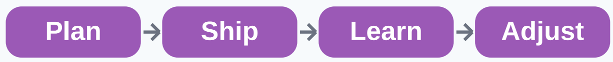

Figure 1. Each loop generates evidence; pure planning loops generate confidence without it.

HOW IT IS USED

Set a default: when a decision has been analysed for more than X hours/days without clear superiority of any option, ship the best current option and learn from feedback. Engineer rituals that lower the cost of action and raise the cost of further analysis.

WHEN IT IS USED

Use whenever analysis is producing diminishing returns, in agile and lean contexts, in venture-capital portfolio construction, and as a counterweight to perfectionist team cultures.

BY WHOM IT IS USED

Founders, agile coaches, accelerator directors, product managers, and senior leaders steering organisational culture toward execution speed.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where classical strategy (Andrews, 1971) prized comprehensive analysis and Mintzberg (1978) described emergent strategy descriptively, Peters and Waterman's contribution is the prescriptive default-to-action operating principle. Building on Toyota's *genchi genbutsu*, it codifies action-bias as a culture-design lever rather than personal trait.

RELATED FRAMEWORKS IN THIS COURSE

Lemonade · Slibrary 10
Build-Measure-Learn · Slibrary 15
Riskiest Assumption Test · Slibrary 15
Pre-Mortem · Slibrary 11

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Growth vs Fixed Mindset

SLIBRARY 12 · SHEET 4 / 5

Dweck — ability is malleable, not fixed

PROFESSIONAL DEFINITION

Carol Dweck's foundational research distinguishes *growth mindset* (the belief that ability is malleable through effort and learning) from *fixed mindset* (the belief that ability is innate and unchangeable). Mindset shapes response to challenge, persistence after failure, and openness to feedback — the variables most predictive of entrepreneurial venture survival. Mindset is itself malleable: targeted interventions can shift individuals from fixed toward growth (Dweck, 2006).

WHO DEVELOPED IT

Carol S. Dweck, Stanford psychology professor, in *Mindset: The New Psychology of Success* (2006), synthesising thirty years of empirical research starting with her work on learned helplessness in children (Dweck, 1975). The work has been translated into education, sport, parenting and organisational contexts.

WHY IT WAS DEVELOPED

Performance differences across individuals were attributed to fixed traits (talent, IQ). Dweck's research showed that beliefs about ability — independent of actual ability — predicted persistence and learning curves. Identifying mindset as malleable made performance gaps interventionally addressable.

FIGURE 1 · TWO MINDSETS

Fixed	Growth
Avoid challenge	Embrace challenge
Give up	Persist
Effort futile	Effort works
Ignore feedback	Learn from feedback

Figure 1. Mindset shapes response to challenge, feedback and failure — the variables most predictive of venture survival.

HOW IT IS USED

Audit team members and self for fixed-mindset signals (avoiding hard tasks, defensive response to feedback). Reframe challenges as learning opportunities. Praise effort and strategy over outcomes. Build feedback rituals that normalise mistakes as data.

WHEN IT IS USED

Use in onboarding, in performance-development conversations, in leadership coaching, in early-stage venture-team selection, and in MBA-curriculum design for entrepreneurship.

BY WHOM IT IS USED

Educators, coaches, HR leaders, venture-capital investors evaluating founder coachability, and individual professionals self-diagnosing performance plateaus.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Bandura (1977) introduced self-efficacy and McClelland (1961) studied need for achievement, Dweck's contribution is the binary mindset typology with rigorous interventional evidence. Building on attribution theory (Weiner, 1972), it gave performance psychology a teachable, action-oriented frame.

RELATED FRAMEWORKS IN THIS COURSE

- Kelley's Creative Confidence · Slibrary 2
- Dosi's DT Mindset Measure · Slibrary 2
- Bias for Action · Slibrary 12
- Entrepreneurial Orientation · Slibrary 12

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Entrepreneurial Orientation

SLIBRARY 12 · SHEET 5 / 5

Innovativeness · Proactiveness · Risk-taking · Autonomy · Competitive aggression

PROFESSIONAL DEFINITION

Entrepreneurial Orientation (EO) is a psychometrically validated five-dimension construct measuring an organisation's strategic posture: *innovativeness* (pursuit of novel solutions), *proactiveness* (anticipating market changes), *risk-taking* (willingness to commit to uncertain opportunities), *autonomy* (independent action by champions), and *competitive aggressiveness*. EO scales correlate consistently with firm growth across industries and decades (Lumpkin and Dess, 1996; Covin and Slevin, 1991).

WHO DEVELOPED IT

Jeffrey Covin and Dennis Slevin developed the original three-dimension measure (1991) building on Miller (1983). Lumpkin and Dess (1996) extended to five dimensions in their *Academy of Management Review* paper, which became the dominant operationalisation.

WHY IT WAS DEVELOPED

Practitioners debated which firm-level traits drove growth, but without measurement the debate was unresolvable. Covin and Slevin's instrument made entrepreneurial posture a quantitative variable researchers could correlate with outcomes — and Lumpkin and Dess's extension captured nuance the original missed.

FIGURE 1 · FIVE EO DIMENSIONS

Figure 1. Five independently varying dimensions; profiles diagnose the specific entrepreneurial posture the firm embodies.

HOW IT IS USED

Measure organisational EO via the validated questionnaire (typically 9–15 items per dimension). Aggregate scores show overall posture; dimension scores reveal which specific strategic stance the firm currently embodies.

WHEN IT IS USED

Use in corporate-entrepreneurship audits, in organisational-development diagnoses, in pre-acquisition cultural-fit assessments, and in academic research correlating posture with outcomes.

BY WHOM IT IS USED

Strategy-research academics, corporate-strategy teams measuring organisational entrepreneurial capacity, HR analytics teams, and consultants advising on entrepreneurial-culture transformation.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Miller (1983) introduced the entrepreneurial-firm concept and Covin and Slevin (1991) gave it three dimensions, Lumpkin and Dess's (1996) contribution is the five-dimension extension that decomposed posture into independently varying attributes. Building on Mintzberg's (1973) entrepreneurial-mode work, it gave EO theoretical specificity.

RELATED FRAMEWORKS IN THIS COURSE

- Growth vs Fixed Mindset · Slibrary 12
- Bias for Action · Slibrary 12
- Causal vs Effectual · Slibrary 12
- Ambidextrous Organisation · Slibrary 20

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